

ssh Cheat Sheet

Log in to another machine securely, run commands on it, copy files, and tunnel traffic through it.

Using it

```
ssh [options] user@host [command] ← no user = your local username · no command = interactive shell
ssh mark@203.0.113.10 'df -h' ← runs one command on the server and returns
```

Option	Means	Option	Means
-p 2222	Connect to a different port	-N	No shell; just hold the tunnels open
-i key	Use this private key file	-f	Go to the background after login
-v / -vvv	Show what's happening (debug)	-t	Force a terminal (for sudo , top)
-J jump	Hop through another host first	-C	Compress (helps on slow links)

Keys: log in without a password

Command	Does
ssh-keygen -t ed25519 -C 'mark@laptop'	Makes a key pair: <code>~/.ssh/id_ed25519</code> (secret) and <code>.pub</code> (shareable)
ssh-copy-id mark@203.0.113.10	Adds your public key to the server's <code>~/.ssh/authorized_keys</code>

~/.ssh/config: nicknames and defaults

```
Host web1 ← nickname: now just type ssh web1
  HostName 203.0.113.10
  User mark
Host db1
  HostName 192.168.1.20
  ProxyJump web1 ← reach db1 by hopping through web1 (same as ssh -J web1 192.168.1.20)
```

Port forwarding (tunnels)

```
ssh -L 8080:localhost:80 mark@web1 ← Local: your localhost:8080 reaches port 80 on web1
ssh -L 5432:192.168.1.20:5432 mark@web1 ← Local: reach a machine behind web1 as if it were here
ssh -R 9000:localhost:3000 mark@web1 ← Remote: web1's port 9000 reaches port 3000 on your machine
ssh -D 1080 mark@web1 ← Dynamic: SOCKS proxy; browser traffic exits from web1
ssh -N -L [listen port]:[host]:[port] server ← host is as seen from the server · -N = forward only, no shell
```

Copying files

```
scp report.txt mark@web1:/tmp/ ← upload
scp mark@web1:/var/log/syslog . ← download into the current folder
scp -r -P 2222 mydir mark@web1: ← a whole folder, into your home folder; port is a capital -P
sftp web1 ← interactive: ls, cd, get file, put file, lcd (local cd), bye
```

Permissions (ssh ignores keys others could read)

```
chmod 700 ~/.ssh ← only you can open the folder
chmod 600 ~/.ssh/id_ed25519 ~/.ssh/authorized_keys ← only you can read these (.pub files can be 644)
```

When it goes wrong

You see	Means / fix
Permission denied (publickey)	Key refused. ssh -v shows keys tried; check permissions on both ends
Connection refused / timed out	Refused = sshd not running or wrong port. Timeout = wrong IP or firewall
IDENTIFICATION HAS CHANGED	Server key changed (rebuild or attack?). If expected: ssh-keygen -R web1
UNPROTECTED PRIVATE KEY FILE!	Others can read your key: fix with the chmod lines above
Session frozen	Press Enter then ~. to kill it (~? lists other escapes)

Remember: the **private** key never leaves your machine; only the **.pub** goes on servers. Before disabling password login, keep one session open and test the key in a second one.